

CSCE 640 QUANTUM ALGORITHMS · MODULE 05 · LECTURE NOTES

The Query Model

Where the separations are provable, and what that is worth.

MODULE OUTCOMES

- Explain the query model and why it is used.
- Derive the Deutsch-Jozsa algorithm.
- Explain Simon's problem as the bridge to Shor.
- Distinguish query complexity from time complexity.
- State what a query separation does and does not establish.

1 The model

IN THE QUERY MODEL YOU COUNT ORACLE CALLS AND IGNORE EVERYTHING ELSE, WHICH IS WHAT MAKES LOWER BOUNDS PROVABLE

The input is a function you may evaluate but not inspect — so the only resource counted is the number of evaluations, and all other computation, however much, is free.

Which makes lower bounds tractable: you can prove that k queries cannot possibly distinguish two cases, because **that is a statement about information rather than about circuits** — and circuit lower bounds are notoriously out of reach (CSCE 637 Module 09).

And that is precisely why this subject's provable separations are nearly all query separations — Deutsch-Jozsa, Simon, Bernstein-Vazirani, and Grover's optimality — while the result people actually care about, factoring, is not proven at all (Module 12 §3).

So the model is a tool for making honest statements — and the honest statement about it, in §4, is that a query separation is weaker than it sounds, which does not stop it from being a real theorem.

2 Deutsch-Jozsa

PROBLEM

$f: \{0,1\}^n \rightarrow \{0,1\}$ is promised to be either constant or balanced. Which one?

CLASSICAL

$2^{n-1} + 1$ queries in the worst case, deterministically. (Randomised: $O(1)$ for high confidence -- which matters; see the callout.)

QUANTUM, one query

- 1 H on all n qubits: uniform superposition
- 2 phase oracle: $|x\rangle \rightarrow (-1)^{f(x)}|x\rangle$
- 3 H on all n qubits again
- 4 measure

WHY IT WORKS

amplitude of $|0\dots 0\rangle$ after step 3 is
 $(1/2^n) \cdot \sum_x (-1)^{f(x)}$
 constant $f \Rightarrow$ that sum is ± 1 , so we measure all zeros with certainty
 balanced $f \Rightarrow$ the sum is exactly 0: perfect cancellation, so all zeros is impossible

Constant f reinforces on the all-zeros outcome and balanced f cancels there exactly — which is Module 04's mechanism in its clearest possible form: the two cases are distinguished by whether a particular sum of ± 1 terms is extremal or zero. Trace this fully by hand; it is the model example of the whole course, and the second layer of H gates is where the path sum is actually performed.

AND THE HONEST CAVEAT, WHICH IS THE REASON THIS IS A TEACHING EXAMPLE AND NOT A RESULT

The exponential gap is against *deterministic* classical algorithms — a randomised classical algorithm answers with high confidence in a constant number of queries, simply by sampling a few points and seeing whether they agree.

So the dramatic separation requires exact classical computation, which is an unusual thing to demand — and stating it as 'exponentially faster' without that qualification is misleading, though it is frequently done in popular accounts.

Which is why Simon's problem matters considerably more — it is an exponential separation against *randomised* classical algorithms, which is the meaningful comparison since nobody is restricted to deterministic methods (CSCE 658's whole premise).

And Simon's structure is period-finding — a hidden XOR mask, located by interference — which makes it the direct ancestor of Shor's algorithm (Module 08), and is why it is worth more attention than its obscurity suggests.

3 Simon's problem

promise: $f(x) = f(y)$ iff $y = x \oplus s$, for a hidden nonzero s

- **f is two-to-one with a hidden XOR period s** , and the task is to find s — a promise problem, like Deutsch-Jozsa, but with an answer to extract rather than a bit to decide.
- **Classically it takes $\Omega(2^{n/2})$ queries, even with randomisation** — because you must find a collision, and collisions are rare until you have seen about the square root of the domain (a birthday argument, CSCE 658 Module 04). **This is the meaningful lower bound**, and it is the one Deutsch-Jozsa lacks.
- **Quantumly it takes $O(n)$ queries, plus classical linear algebra: each run yields a uniformly random y satisfying $y \cdot s = 0 \pmod{2}$** — the interference kills every y that does not satisfy it — and $n - 1$ independent such constraints determine s uniquely.
- **Each quantum run returns a random linear constraint on s , and n of them determine it** — which is exactly the same shape as Shor's algorithm: a quantum subroutine that produces one random piece of information about a hidden period, called repeatedly, finished classically.
- **Simon is the bridge: hidden periodicity, found by interference, then finished classically** — and recognising that shape is what makes Module 08 feel inevitable rather than magical.

4 What it establishes

- **A query lower bound is a real theorem, and it is unconditional** — unlike almost everything else in complexity theory, which rests on unproven assumptions (CSCE 637 Module 01's framing). That is a genuine and unusual strength.
- **But it is relative to an oracle**, and **an oracle separation does not imply a separation for explicitly given functions** — which is a known and important gap, and the reason oracle results cannot settle questions like P versus NP (CSCE 637 Module 10's relativisation barrier).
- **And it ignores the oracle's own implementation cost** (Module 03 §3) — so an $O(1)$ -query algorithm that calls a circuit of exponential size is not a fast algorithm, and the query count says nothing about that.
- **Which is the trap: 'exponentially fewer queries' is routinely reported as 'exponentially faster', and they are different claims** — the substitution appears in abstracts, press releases, and occasionally in textbooks.
- **So the model proves real things, and the things it proves are narrower than the headlines** — both halves of which are worth holding at once, which is the general posture this course asks for.

AND WHAT TO CARRY FORWARD

The query model is where this subject's honest theorems live, and Grover's optimality (Module 06 §4) is the most important of them — it bounds what unstructured search can ever buy, for all time, which is a rare kind of knowledge.

Simon's structure is the one to remember — hidden periodicity, extracted by interference, finished by classical linear algebra — because Shor's algorithm has exactly that shape with the XOR group replaced by the integers mod N .

And the discipline is to state which complexity you are talking about — queries, gate count, depth, T-count, or wall-clock — which is Module 03 §4's five metrics restated as an honesty requirement rather than an engineering one.

Which is this module's version of the program's rule: name the resource you counted, because the separation is a statement about that resource and not about speed in general.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Deutsch & Jozsa — Rapid solution of problems by quantum computation	§2 in the original — short, and the promise structure is stated clearly.
Simon — On the power of quantum computation	§3 in the original — and the paper Shor credits as the one that suggested the approach.
Watrous's notes on query complexity (free)	§§1 and 4 — the model stated rigorously, with the lower-bound methods.
Beals et al. — Quantum lower bounds by polynomials (free)	§4's technique — the polynomial method, which is how most query lower bounds including Grover's are proved.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 State the query model** and say why lower bounds are easier in it.
- 2 Trace Deutsch-Jozsa for $n = 2$** , all four f , by hand.
- 3 Compute the all-zeros amplitude** for a balanced f and verify it is zero.
- 4 Implement it** and confirm the certainty of both cases.
- 5 Give the randomised classical algorithm** and its query count.

- 6 **State Simon's promise** and the classical lower bound's argument.
- 7 **Implement Simon's algorithm** for $n = 4$ and recover s .
- 8 **Count the runs needed** and compare to $n - 1$.
- 9 **Explain the relativisation gap** in your own words.
- 10 **Find a paper or article** that substitutes 'faster' for 'fewer queries'.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What is counted in the query model, and why use it?
- 2 Why are query lower bounds provable when circuit ones are not?
- 3 State Deutsch-Jozsa's promise and the quantum algorithm.
- 4 What cancels, and in which case?
- 5 State the caveat about the classical comparison.
- 6 Why does Simon's problem matter more?
- 7 Give Simon's promise, both bounds, and the mechanism.
- 8 What shape does Simon share with Shor?
- 9 Name three limits of a query separation.
- 10 What is the routine substitution to watch for?